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Oil Theft and Violence in Mexico*

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Abstract

Around the world, non-state armed actors have been linked to the illegal extraction of energy resources. This research note explores the case of Mexico. Anecdotal evidence suggests that criminal groups have been gaining control of energy infrastructure across the country. At the same time, oil tapping has been directly associated with criminal violence. Yet, there has not been a systematic effort to causally identify the relationship between illegal extraction and criminal violence. In this research note, we use the exogenous variation in international oil prices—as a measure of potential profits—to assess the effect of access to energy infrastructure on criminal-related violence. Our results show that increases in oil prices are associated with higher levels of homicide rates in municipalities with pipelines and in neighboring municipalities. Specifically, a standard price increase during this period is associated with approximately 20% more homicides per year in municipalities with gasoline pipelines. A locality-level analysis suggests a non-linear distance effect. Finally, we also explore criminal fragmentation as the

*The data and code to replicate the findings of this study are available at: Martinez-Alvarez, Cesar Benshuni, 2023, "Replication Data for: "Oil Theft and Violence in Mexico", https://doi.org/10.7910/DVN/YIY3B5, Harvard Dataverse, V1.

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mechanism connecting access to resources and violence. We find that access to pipelines is associated with higher presence of organized crime groups, but not necessarily with more fragmentation.

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Keywords
criminal violence, oil theft, natural resources, Mexico

Introduction

During 2018, the middle-size city of Irapuato in the state of Guanajuato, central Mexico, became the fourth most violent city in the world.[1] In that year, the state became a hotspot of violence due to confrontations between the dominant Jalisco New Generation Cartel (*Cartel Jalisco Nueva Generación*) and the newcomer *Santa Rosa de Lima* Cartel (Corcoran, 2019). These groups were not fighting for controlling a trade route or a turf related to the profitable drug market in the United States. Instead, violence erupted to seize control of the local oil theft business, a booming activity pursued by Organized Crime Groups (OCGs).

In 2018, PEMEX, the Mexican state-owned oil company, reported 12,581 cases of fuel theft involving pipe tampering (a practice known as *huachicol*), a 20% increase from the previous year.[2] Although there is a plethora of anecdotal evidence pointing to a direct relationship between the presence of gasoline pipelines, oil theft, and the rise of violence in Mexico, there has not been a systematic effort to identify a causal link.[3] In this paper, we address this gap in the literature.

Measuring a causal relationship between oil theft and violence in the Mexican case is not straightforward. First, as shown in Figure 1, violence in Mexico has been increasing for a long time before criminal organizations adopted oil theft as part of their criminal portfolio. This means that there may be many other reasons unrelated to oil theft that could explain the increase in violence (Flores Martínez and Phillips, 2022). An approach to address this problem would be to compare violence in places with access to pipelines versus places without them. A problem with this methodology is that, even if violence significantly increased at higher rates in places with access to pipelines, this change might be still correlated to other factors also affecting violence, such as state presence or the value of a given municipality in the drug trade.[4]

To causally identify the relationship between pipeline presence and violence we take advantage of Mexico's position as a price-taking economy in the global petroleum markets.[5] We focus on how changes to global oil prices impact violence in municipalities and localities with access to gasoline pipelines. The expectation is that, when international prices increase and this commodity becomes more valuable, OCGs would have incentives to diversify their portfolio by capturing revenues from this commodity. The proposed mechanism connecting this behavior with violence is cartel competition and fragmentation. The main advantage of our empirical strategy is that, because they

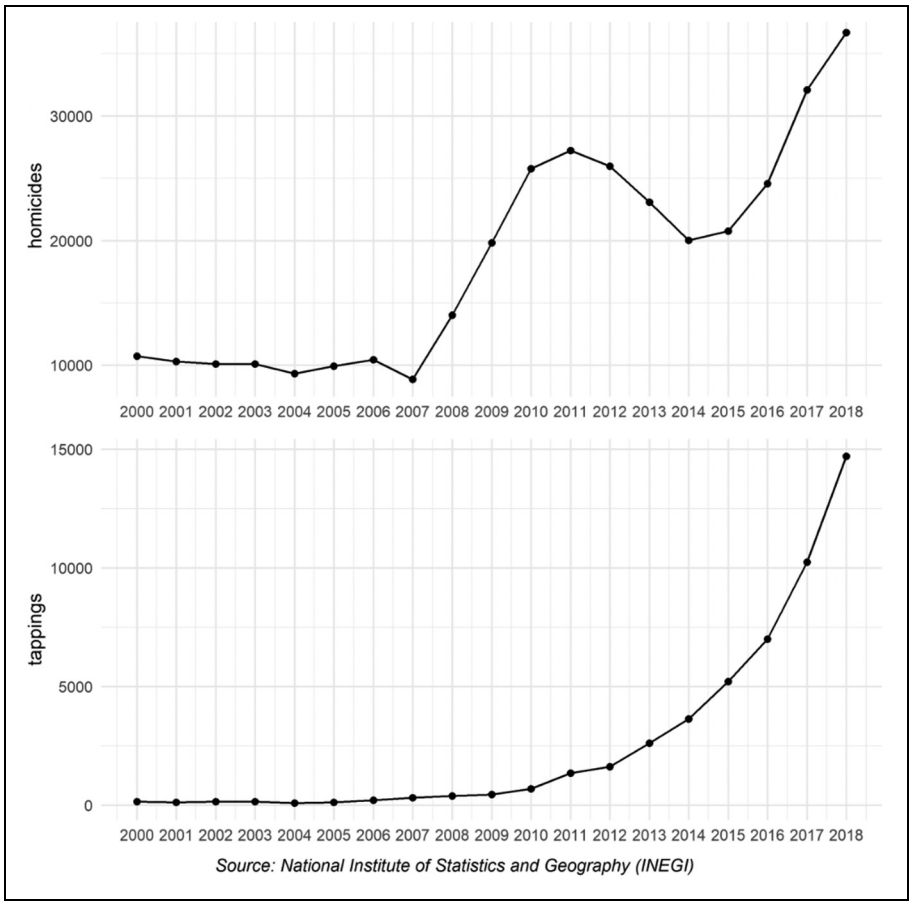

Figure 1. Homicides and Illegal Tappings in Mexico.

are plausibly exogenous, international prices are not correlated with other variables that influence the homicide rate.

Our results show that a standard increase in the global crude oil prices (equivalent to approximately USD 30 per oil barrel) is associated with 1.65 additional murders per 100,000 per year in municipalities with fuel pipelines. To put this figure in context, the mean homicide rate at the municipality level in our period of analysis is 7.86.[6] We also provide evidence that oil-related violence affects neighboring municipalities—hence, high oil prices have a spatial spillover effect.

To gain a deeper understanding of this relationship, we then turn to a locality level analysis. Here, we compare the effects of the exogenous shock in areas located at different distances from a fuel pipeline. We assume that the cost of transporting and selling oil products is lower for localities close to the pipeline network; hence, the incentives to access these

resources are higher. Our results confirm that distance to a pipeline is the main determinant of the violence increase when oil prices rise. However, the effect seems to be stronger in localities within a "medium" distance to this infrastructure (between 15 and 20 km).[7]

We provide a test of our proposed mechanisms by using data on criminal organization presence at municipal level. Our results show that access to pipelines is correlated with a higher presence of criminal organizations, which is also associated with more violence. However, higher prices do not seem to increase the number of existing criminal groups. These results suggest that violence associated with these resources occurs because of competition between existing groups, not necessarily because of the emergence of new ones.[8]

We make contributions to two different strands of research.[9] First, we engage directly with the literature on organized crime. Scholars have analyzed how criminal organizations expand their enterprises beyond drug trafficking, particularly by extorting citizens (Gambetta, 1996; Magaloni et al., 2020; Moncada, 2013) and interfering in the political process (Trejo and Ley, 2020). However, there is less empirical evidence on whether and how criminal organizations diversify their activities to other markets, including natural resources (Herrera and Martinez-Alvarez, 2022). Although there is a consensus that—when commodities are easily captured—price booms can increase violence (Blair et al., 2021; Dube and Vargas, 2013; Sánchez De La Sierra, 2020), this research has mainly focused on rebel groups. Our analysis suggests that these dynamics replicate in a context where criminal organizations operate.

This paper also contributes to the "resource curse" literature (Ross, 2001, 2006), specifically, to the hypothesis that links resource extraction and violence (Fearon, 2005; Ross, 2004). The literature identifies two potential mechanisms connecting these variables: the access of rebel groups to these resources and the lack of state capacity (Fearon, 2005; Fearon and Laitin, 2003; Snyder and Bhavnani, 2005). Until recently, most of these studies were based on cross-national data, which limited their ability to disentangle the mechanisms (Hong and Yang, 2020). Nonetheless, an emerging area of research at the subnational level has improved our theoretical understanding of the reasons why natural resources exacerbate conflict (Haass, 2021; Sanchez de la Sierra, 2020; Sexton, 2020). Our study contributes to this literature by showing that competition between criminal organizations to obtain access to these resources is a key factor to explain increases in violence.[10]

Background

Mexico represents an ideal case to study the relationship between access to oil revenues and criminal violence for two reasons. First, the increasing competition among criminal organizations and the state-led efforts to eradicate them have incentivized these groups to look for additional sources of revenues. Second, gasoline and other oil products are relatively easy to capture, offering an attractive and accessible source of profits.

Criminal Violence and Revenue Diversification

Although criminal organizations have operated in the country for decades, recent clashes among groups and with state forces have been associated with rising levels of criminal

violence (Astorga, 2016; Castillo and Kronick, 2020; Herrera, 2019). The acute increase in homicide rates in Mexico since 2006 is a complex phenomenon with multiple causes. These include the disruption of drug-trade routes (Dell, 2015), easing the acquisition and smuggling of weapons (Dube, Dube, and Garcia Ponce, 2013), and changes in drug consumption preferences (Heinze and Armas Castañeda, 2016). Politically, according to Trejo and Ley (2020) the democratization process, in particular at the sub-national level, disrupted the protection networks that criminals and politicians associated with the PRI (Partido Revolucionario Institutional in Spanish) had built over decades.

The state's efforts to eradicate drug trafficking are another major cause in the most recent wave of homicidal violence in Mexico. First, the launching of the War on Drugs in late 2006 by President Calderon led to the disruption of criminal networks. As several scholars have argued (Calderón et al., 2015; Flores Martínez and Phillips, 2022; Osorio, 2015; Rios, 2013), the use of state forces to eradicate criminal organizations, for example, by beheading these groups (Calderón et al., 2015; Phillips, 2015), usually leads to more intra-group competition among smaller units, which, in turn, leads to higher levels of lethal violence. Second, as Dube and Vargas (2013) show, drug seizures are also associated with higher levels of violence, as the main profit source for these groups shrinks.

As a result, the competition to seize revenues from the drug market has increased. This, in turn, has incentivized some criminal organizations to diversify their portfolios. Although there are multiple journalistic accounts about this practice (Lemus, 2018), scholarly evidence is relatively scarce. Nonetheless, there is some quantitative and qualitative evidence that criminal groups have targeted sectors other than the drug industry, including extortion to small businesses, and mining and agriculture (Herrera and Martinez-Alvarez, 2022; Magaloni et al., 2020; Ornelas, 2018).

Natural resources are a particularly attractive source of additional revenues. Oil products, in particular gasoline, differ from other commodities for several reasons. First, in contrast to the mining and export-agriculture industries, which tend to be very concentrated in a small number of municipalities in Mexico (Herrera and Martinez-Alvarez, 2022), gasoline is an ubiquitous product available across the country. As such, it is much more accessible than other valuable commodities. Second, processed fuel is much easier to sell in illegal markets than other goods, which makes it a very attractive source of economic diversification (León Sáez, 2021a). Finally, the lack of investment in Mexico's midstream oil infrastructure has facilitated the extraction of gasoline by criminal organizations (León Sáez, 2021c).

Oil Infrastructure and Theft

The structure of the Mexican oil industry is another reason why criminal groups have pivoted toward the natural resources sector. The country has been among the top producers of crude oil for decades—extraction first began in the early twentieth century (Haber et al., 2003; Uhthoff López, 2010).

Although output has decreased considerably since the 2003 peak and the federal budget has diversified away from oil, the fossil fuel industry is still relevant in economic

and political terms (Cunningham, 2019; López Pérez and Vence, 2021; Rosellon and Zenón, 2012).[11] In terms of demand, Mexico's consumption of refined oil ranks among the largest in the developing world in per capita terms. As a result, the country has an extensive network of crude oil, gasoline, and natural gas pipelines (Torres and Garza, 2015; Uhthoff López, 2010).

This network has three main characteristics. First, although the crude oil pipelines concentrate in the southeast of the country, the gasoline midstream infrastructure is widely distributed across Mexico, connecting the largest urban areas. Second, the vast majority of these pipelines were built from the 1930s to the early 2000s by the National Oil Company—that is, before the onset of the War on Drugs (Torres and Garza, 2015; Uhthoff López, 2010).[12] Finally, the dimensions of the network and the relative lack of recent investment have made it quite vulnerable to attacks by criminal organizations (León Sáez, 2021c; Rodríguez Padilla, 2018; Romo, 2016).

According to journalistic accounts, oil theft has expanded considerably after 2015, in particular by smaller criminal organizations. These groups tend to engage in oil theft as they lack enough capabilities to compete with larger ones for access to other illegal activities. Moreover, as some journalists have pointed out, PEMEX staff have provided inside information and support to these criminal syndicates, facilitating the theft of oil (Dittmar, 2017). The proliferation of smaller organizations exacerbates competition. Although there are reports of oil theft in 26 of the 32 states of the country, the majority of the associated violent events concentrate in central Mexico (León Sáez, 2021b).[13]

Identification Strategy and Data

We seek to estimate the causal relationship between access to refined petroleum resources and criminal violence in Mexico. The identification of such an effect is difficult for several reasons. First, places with high levels of natural resources revenues are different from those without them for reasons correlated with the outcome— homicide rates. The resource curse literature shows that oil-rich countries are, on average, more authoritarian, have higher levels of corruption, and are more likely to initiate civil wars (Ross, 2001).[14] At the subnational level, some authors have found similar differences between oil and non-oil regions (Dube and Vargas, 2013). Hence, there are reasons to assume that political and economic variables are important confounders in the relationship between access to oil-related revenues and homicide rates.

To address these concerns, our identification strategy is based on two sources of quasi-exogenous variation. First, geographically, we employ the location of gasoline pipelines as a proxy variable for access to revenues from the oil sector. We do so at two levels of analysis. First, we divide our municipal sample into four categories: (1) a *treated* group, which has one or more gasoline pipelines, (2) a group of neighboring municipalities without pipelines (to test for spillover effects on violence), (3) a group including the rest of municipalities in the country, and (4) a group of municipalities without pipelines and with high levels of violence.[15] See Figure 2 and the Supplemental materials.

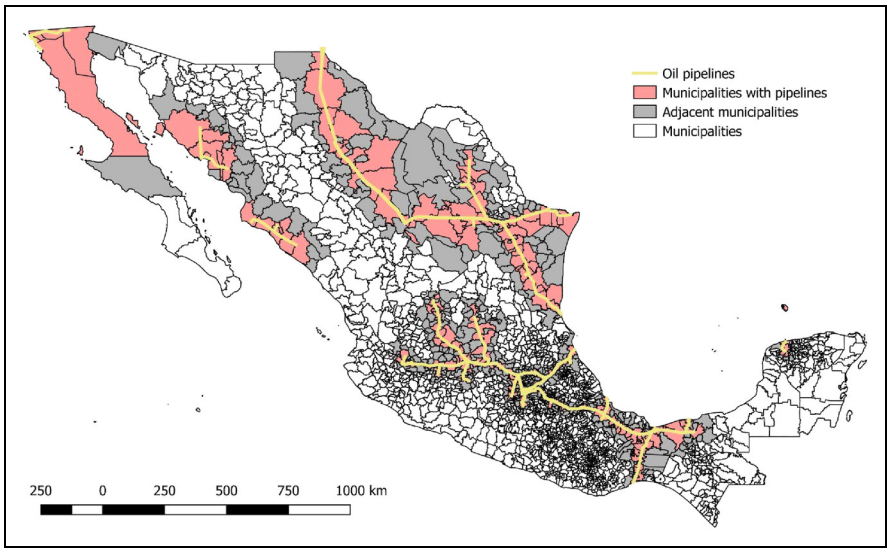

Figure 2. Gasoline Pipelines. Geographic Distribution of Municipalities with Gasoline Pipelines. *Source*: Mexico's Ministry of Energy.

Second, at the locality level, we measure the distance in kilometers from a given locality to the closest fuel pipeline. The rationale behind this variable is that, conditioned on pipeline presence within the municipality, the closer a locality is to energy infrastructure, the more likely it is to become a target for criminal organizations. Because the effect in the presence of pipelines may decrease with distance, we focus on localities within a 30 km buffer area.[16]

The location of energy infrastructure in Mexico is plausibly exogenous to the levels of criminal violence. As mentioned above, the vast majority of the gasoline pipelines were built well before the mid-2000s, when homicide rates increased rapidly. Therefore, it is unlikely that drug-trafficking violence influenced the planning of the midstream sector of the Mexican oil industry. Instead, the main goal of the energy policy makers was to minimize the costs of transportation of oil products from the point of processing to the main consumption centers (Uhthoff López, 2010). Finally, given that gasoline pipelines have few (if any) economic spillovers along the areas where they are located, it is highly unlikely that population centers emerged in response to the location of this network.

The second element of our identification strategy introduces exogenous temporal variation in the value of oil products; specifically, we interact the previous variables (presence and distance to a pipeline) with the international price of crude oil. Because the market is concentrated in a handful of producers, very few of them are capable of swinging the global price of this commodity (Saudi Arabia and Russia, for example). Given that Mexico produces only a small fraction of the international output (approximately 2% in 2019),[17] changes in domestic production (including those related to criminal violence) are quite unlikely to affect global markets. In other words, Mexico is a price taking

economy. The main assumption is that the international price of crude oil is exogenous to Mexico's homicide rates.

Methods and Data

In this paper, we employ two main units of analysis: the municipality and the locality. The main outcome variable in both cases is the number of homicides for men 19–39 years old, normalized by population. Measuring the levels of criminal violence is quite challenging, due to the lack of accurate reporting and the difficulty to attribute a murder to a criminal organization. We follow Calderón et al. (2015) and approximate this variable with the murder rates for young men, as this demographic group is the most likely to experience criminal violence.

We obtained this data from the official mortality statistics of the federal Ministry of Health of Mexico,[18] which have full cross-sectional and temporal coverage for all the country from 1990 to 2017. Homicides at the municipality level are normalized by 100,000 inhabitants to account for different municipal populations. We also include other types of criminal activities that had been associated with OCG presence, specifically extortion, kidnapping, and theft (all normalized per 100,000 inhabitants), using publicly available data from the Executive Secretariat of the National System of Public Security.

An issue of using homicides at locality level is that this variable is highly skewed both geographically and temporally. That is, a handful of localities within a given municipality concentrate the majority of homicides. At the same time, there are several months in which the homicide count is zero. To avoid a problem of zero-inflation and to keep interpretation simple, we opted for transforming homicides at locality level into a binary variable. This variable takes the value of one when a locality has a number of homicides above the mean within a given municipality.[19]

The main independent variable in both the municipality and locality-level analyses is the interaction between the presence of a gasoline pipeline and the international price of crude oil. The data on global oil prices in USD comes from the U.S. Energy Information Administration.[20] The data on the midstream oil industry in Mexico comes from the federal Ministry of Energy.[21] For the municipal-level analysis, we determined whether a municipality has any gasoline pipelines passing through its surface to create treatment and control groups.

For the locality-level analysis, we use the distance in kilometers from a locality to the closest gasoline pipeline. Once we restrict our sample to municipalities and localities with pipeline presence, we end up with 285 municipalities and 23,012 localities. As robustness checks for the municipal-level analysis, we measured the same variables but using crude oil and natural gas pipelines instead of processed fuels, for which we should not observe any effect on the homicide rates[22].

In terms of the estimation, we employ a linear regression model with unit (municipality or locality, depending on the specification) and time (year-months) fixed effects. The inclusion of the unit fixed effects allows us to control for unobserved confounders at the municipality and locality levels. Having time fixed-effects accounts for the relevance of temporal shocks that affect all the units of the sample simultaneously.[23]

Municipality Level Results

Starting with the municipal-level results, Figure 3 shows the coefficients for the main variable of interest—the interaction of pipeline presence and the global oil price—in four different models. In all of them, the outcome is the homicide rate of young men per 100,000. Model 1 compares municipalities with a fuel pipeline with the rest of the country. In Model 2, we include only municipalities with a gasoline pipeline and their neighbors—specifically those without pipelines. Model 3 compares municipalities adjacent to a municipality with a pipeline with the rest of the country—excluding those with that infrastructure. Finally, Model 4 compares municipalities with energy infrastructure with only the most violent municipalities without a fuel pipeline (excluding neighbors of areas with pipelines). The reason to do so is to obtain more comparable samples and reduce the imbalances between places with and without critical gasoline infrastructure.

Three patterns emerge from these models. First, there is a strong association between increases in the international price of petroleum and criminal violence among municipalities with fuel pipelines (Model 1 in Figure 3). The interaction of these two variables is statistically significant at the conventional levels and substantively relevant. An increase equivalent to a standard deviation in the global oil price (USD30) is associated

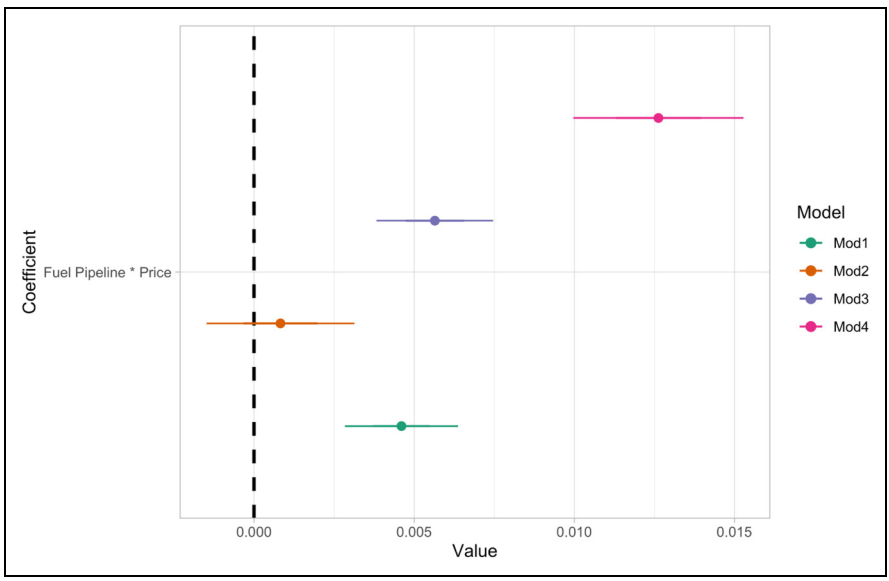

Figure 3. Main Results.
Note: Coefficients for the Interaction Between Prices and Pipeline Presence. Mod 1 Compares Municipalities with Pipeline vs. Municipalities Without Pipelines; Mod 2 Compares Municipalities with Pipelines vs. their Neighbors; Mod 3 Compares Neighbor Municipalities with Municipalities Without Pipelines; Mod 4 Compares Municipalities with Pipelines with Municipalities Without Pipelines with High Levels of Violence.

with 0.138 additional homicides per 100,000 per month or 1.66 per year at municipal level.

Second, changes in the international petroleum price do not affect municipalities with fuel pipelines and their neighbors differently (Model 2 in Figure 3). For these subsamples, the coefficient of the interaction is much smaller and statistically non-distinguishable from zero. Moreover, as Model 3 shows, neighbors of municipalities with a fuel pipeline are not only as violent as those with one, but also have higher homicide rates than the rest of the country (Model 3 in Figure 3). Finally, our results hold if we compare only municipalities with a fuel pipeline with the most historically violent municipalities without this critical energy infrastructure; hence, even for places with similar levels of violence, increases in the relative value of gasoline are associated with more homicides (Model 4).[24]

Heterogeneous Effects and Placebo Tests

Other studies have shown that access to revenues from natural resources have heterogeneous effects along economic and demographic dimensions (Vallejo et al., 2018). We run our main model on different strata of the data, subsetting by population and income. First, Figure 4 shows how the interaction between the global oil price and the presence of fuel pipelines affects municipalities by their population size. Model 1 includes only

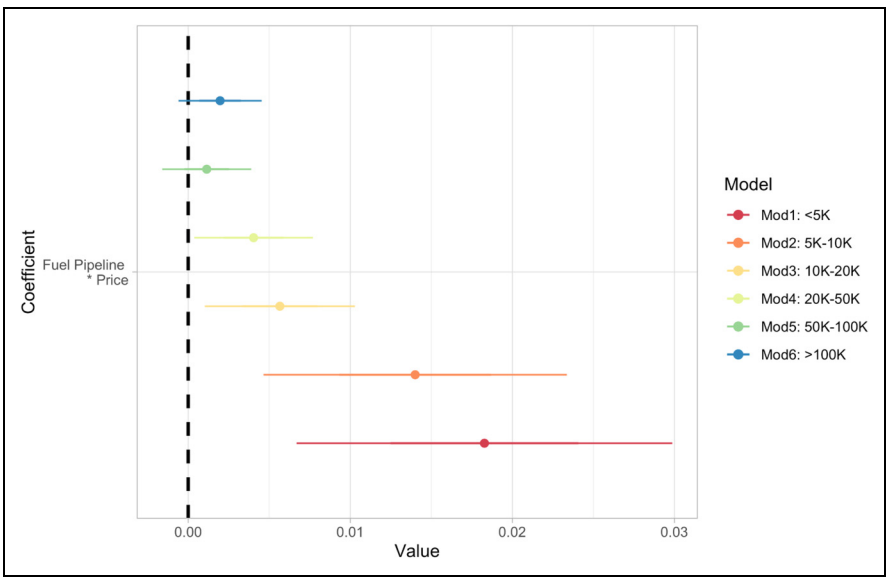

Figure 4. Heterogeneous Effects: Population.
Note: Coefficients for the interaction between prices and pipeline presence for municipalities with different levels of population. The dependent variable is the homicide rate for men from 19 to 39 years old.

municipalities with fewer than 5,000 inhabitants. Model 2 consists of municipalities between 5,001 and 10,000 inhabitants. The sample for Model 3 includes only municipalities between 10,001 and 20,000. Models 4–6 restrict the data to municipalities between 20,001 and 50,000, between 50,001 and 100,000, and more than 100,000 inhabitants.

The findings in Figure 4 show that most of the effect of fuel revenues on criminal violence concentrates in small and very small municipalities. The coefficient is the largest in Model 1 (municipalities with fewer than 5,000 inhabitants); then, the magnitude decreases monotonically as the sample includes bigger municipalities. For the largest municipalities, the coefficient is close to zero and not statistically significant.

Figure 5 shows the heterogeneous effects of oil revenues on criminal violence by levels of income, from the bottom quartile of the distribution (Model 1) to the top quartile (Model 4). Low-medium income municipalities drive most of the impact of higher energy revenues on violence. Finally, we analyzed the relationship between the relative value of gasoline and other variables that should not be affected by it, including different types of criminal activity and other mortality outcomes. We also tested how changes in the global price of oil affect murder rates of municipalities with crude oil and natural gas infrastructure, which, according to our theory, should not be sensitive to this variable. We do not find any strong association across these different models. Supplemental Tables S6–S8 show these results.

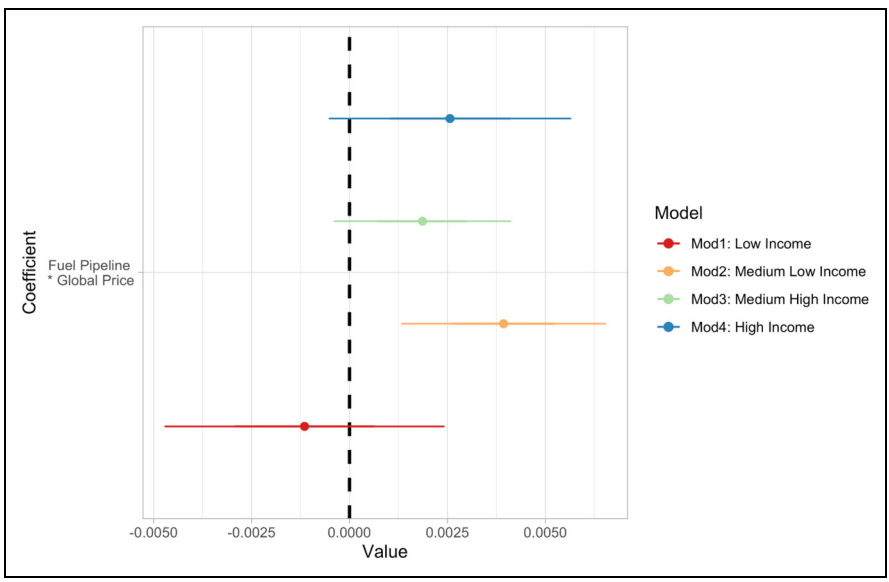

Figure 5. Heterogeneous Effects: Income.
Note: Coefficients for the Interaction Between Prices and Pipeline Presence for Municipalities with Different Levels of Income. The Dependent variable is the Homicide Rate for men from 19 to 39 Years old.

Locality Level Results

In this section, we explore how spatial proximity to a gasoline pipeline impacts the relative levels of violence of thousands of localities in the country. We present these findings in Figures 6 and 7. In each one of the models, the unit of analysis is the locality-year-month. The main dependent variable is a binary variable that takes the value of 1 when a locality has an average murder rate higher than the rest of the localities in the municipality and 0 otherwise. The key independent variable is the interaction between the global price of crude oil and six different versions of *proximity to a pipeline*, a variable which takes the value of 1 when the locality is within a certain distance from the closest pipeline. Figure 6 includes all localities within a 30 km buffer around the gasoline pipeline network of the country. Figure 7 consists only of municipalities with a gasoline pipeline and their immediate neighbors.

There are two main insights from this cut of the analysis. First, the effects of access to energy revenues on violence are not linear with respect to distance. In the two sets of models, changes in the global oil price do not increase the probability of high levels of violence conditional on proximity to a pipeline (5–10 kms). Nonetheless, when we define *proximity* to include all localities within 15, 20, and 25 kms, this variable has a strong and statistically significant positive association with the levels of violence.

Second, the effects of higher global oil prices conditional on proximity to a pipeline are the highest among the municipalities with a neighbor that has a fuel pipeline

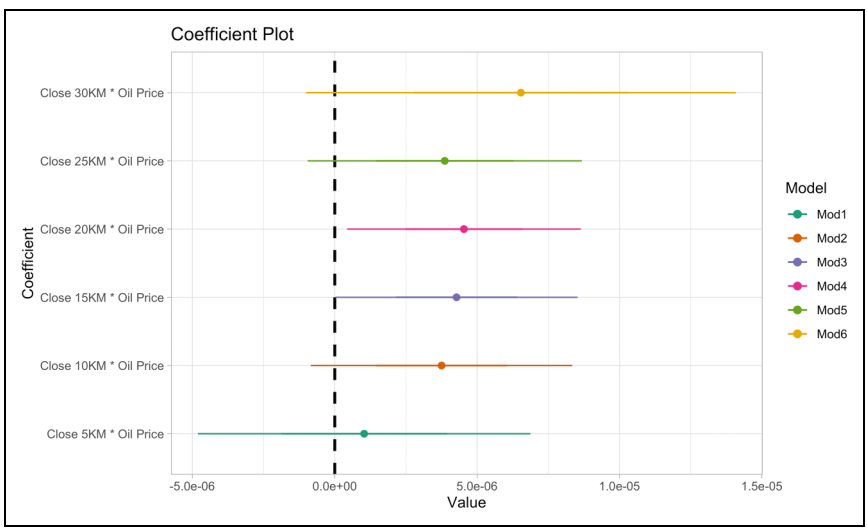

Figure 6. Spatial Spillovers of Oil-Related Violence at the Locality Level: All Municipalities.
Note: Coefficients for the interaction between prices and distance to pipeline for localities within different distances of a pipeline. The dependent variable is the homicide rate for men from 19 to 39 years old.

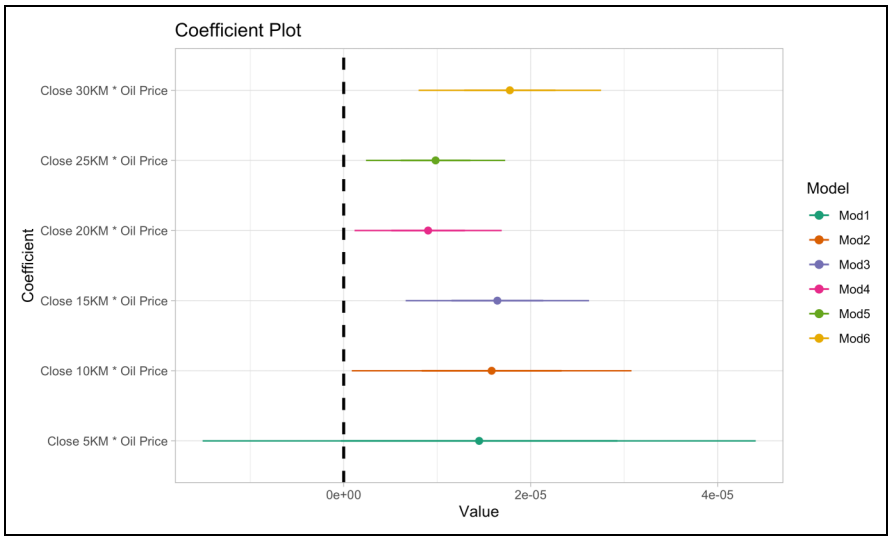

Figure 7. Spatial Spillovers of Oil-Related Violence at the Locality Level: Municipalities with Pipelines and Their Neighbors.
Note: Coefficients for the Interaction Between Prices and Distance to Pipeline for Localities Within Different Distances of a Pipeline in Neighboring Municipalities. The Dependent variable is the Homicide Rate for men from 19 to 39 Years old.

(Figure 7), as the coefficients are four times as large as for the full sample of localities —those located within 30 kms from the pipeline network. Moreover, the interaction is still statistically significant and substantively relevant even for different magnitudes of the distance variable which also include localities farther away from a pipeline. In general, violence levels are not influenced by increases in the price of oil in places immediately adjacent to the fuel network. Instead, violence increases in localities that are farther away from a pipeline (but still relatively close to it), which suggest a strong and substantive effect of oil prices on criminal outcomes in these locations.

Mechanisms

To conclude our empirical section, we present tentative evidence that competition among criminal organizations could be a key mechanism explaining these patterns. However, due to the lack of fine-grained data on the geographic and temporal distribution of these groups, our conclusions remain preliminary. Here, we rely on the CIDE-PPD dataset (Atuesta et al., 2019) as used in Herrera and Martinez-Alvarez (2022), which records the presence of criminal organizations at municipal level for the 2007–2011 period.

First, Figure 8 shows that the presence of a gasoline pipeline is strongly associated with the linear probability of reporting multiple criminal organizations in a municipality

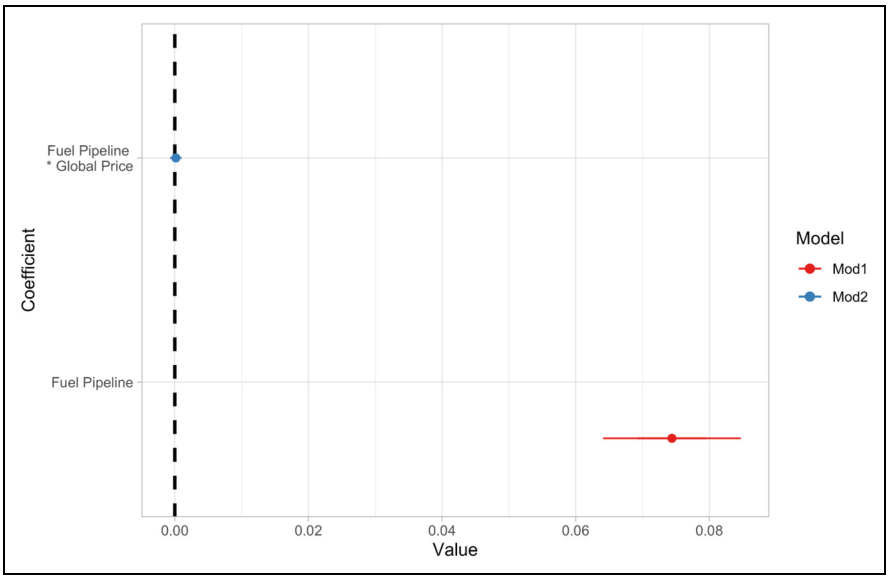

Figure 8. Effect on Number of Criminal Organizations.
Note: Coefficients from the interaction of pipeline presence and oil price (Mod 1) and for pipeline presence (Mod 2). The dependent variable is the number of criminal organizations in a given municipality.

Table 1. Mechanisms.

	Model 1	Model 2
Intercept		0.182134*** (0.010538)
High Criminal Competition	2.131768*** (0.469071)	5.445080*** (1.441371)
Municipality Fixed-Effects	Y	Y
Year-Month Fixed-Effects	Y	Y
Num. obs.	12,270	12,270
Adj. R^2 (full model)	0.703729	0.092337
Num. groups; municipalities	2,454	
Num. groups: year-months	5	

Note: The outcome in both models is the murder rate for men from 19 to 39 years old. The independent variable in both models is a binary variable that takes the value 1 of when the municipality-year reports killings associated with more than one criminal organization and 0 otherwise. Model 1 does not include unit and time fixed effects. Model 2 includes unit and time fixed effects.
$*p < .1$; $**p < .05$; $***p < .01$.

(this variable takes the value of 1 when more than one of these actors are reported in a given municipality-year, according to the CIDE-PPD data and zero otherwise). Nonetheless, changes in the relative value of this commodity (the interaction of pipeline

and global oil price) are not related to the number of actors. These findings are consistent with existing research (Herrera and Martinez-Alvarez, 2022) that finds similar dynamics for the presence and relative value of mining and agricultural commodities in Mexico.[25]

Second, Table 1 presents a model using the homicide rate as the outcome variable and the measure of DTO presence as the main independent variable. There is a positive and strong correlation between having multiple criminal organizations reported in a municipality-year and the overall levels of criminal violence, as generally suggested in the literature (Franco-Vivanco et al., 2020; Magaloni et al., 2020). Although tentative, these findings point to a potential connection between access to gasoline transportation infrastructure, criminal presence, and murder rates in Mexico.

Conclusion

In this research note, we explore the relationship between oil-theft and criminal violence in Mexico. Despite the abundance of anecdotal evidence, this relationship has not been causally identified. Taking advantage of Mexico's position as a price taking economy in the global oil market, we employ exogenous price shocks to measure how they affect the incidence of OCG-related deaths in areas with access to pipelines. Overall, we find that increases in the international oil price are related to larger OCG-related homicides. Specifically, a standard price increase during this period (30 USD per oil barrel) is associated with 20% more homicides per year in municipalities with pipelines. The effect is similar in municipalities that have a pipeline and in their neighbors. This result strongly suggests a spillover effect. The effect is higher in less populated municipalities and in municipalities with low-medium income levels.

Our locality level analysis provides further details about the dynamics of local violence. Specifically, we find that in municipalities with pipelines, violence is concentrated in localities at a "medium" distance from the pipeline (between 15 and 20 km). That is, the effect of distance is not linear. In neighboring municipalities, the effect seems to be concentrated in localities closer to the municipal border. In other words, although the average effect is similar, the local dynamics of violence are different. These results might suggest that OCGs act strategically to avoid higher presence of state forces around pipelines.

Our findings open several opportunities for future research. As the anecdote used in our introduction shows, newcomer cartels are challenging well established drug trafficking organizations to gain access to this profitable market. To disentangle the nature of this conflict, it is necessary to understand which groups are engaging in violence and how oil theft influences cartel diversification. More broadly, our results call to reevaluate the role of geography in the analysis of illegal commodity extraction and violence. As our findings suggest, Mexican OCGs do take advantage of opportunity and weak state presence in a dynamic way. There are reasons to believe that other OCGs around the world, and rebel groups might do the same.

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Supplemental Material

Supplemental material for this article is available online.

Notes

1. According to data from the Mexican non-governmental organization Consejo Ciudadano para la Seguridad Pública y la Justicia Penal (CCSPJP, 2019).
2. According to PEMEX official data (PEMEX, 2022). During 2021, the most recent data available, this figure was 11,037 (ICC, 2022).
3. Some of the few works addressing this topic are: Shuldiner (2022) and Navarro (2018). For a more in-depth analysis see the ICC (2022) briefing.
4. Calderón et al. (2015) identify that certain locations such as proximity to ports, borders, and highway crossroads are high-value turfs, these characteristics could be also correlated with presence of energy infrastructure.
5. An empirical strategy similar to Dube and Vargas (2013) for the Colombian case.
6. The largest increase in the international petroleum price in our period of analysis was USD 79.37, from January 2007 to July 2008.
7. We interpret this result as suggestive evidence that, after tampering with the pipeline, OCGs prefer to operate at a safe distance from this key infrastructure to avoid the higher levels of law enforcement around it.
8. At least during the period for which we have available data on criminal organization presence (2007–2011).
9. Given the nature of this note, our contributions are purely empirical.
10. Another relevant contribution of this paper is our focus on gasoline theft, which is as a source of funding for criminal organizations in many parts of the world (Alonso Berbotto and Chainey, 2021; López-Vallejo and del Pilar Fuerte-Celis, 2021; Onuoha, 2008).
11. According to the Mexican Department of the Treasury (SHCP in Spanish), in 2021, 19.4% of the federal budget revenue came from oil (SHCP, 2021).
12. Official documents from the Mexican senate (permits to transport oil products via pipelines) show that the vast majority of the segments of the gasoline pipeline network were built between 1975 and 2000.
13. According to official PEMEX data obtained via the National Institute for Information Access (INAI), through a freedom of information request, number 1857200094319 from February 2019, available online in the official data repository from INAI.
14. Although the causality between these variables is contested (Haber and Menaldo, 2011).
15. The objective is to reduce baseline differences in the levels of violence between municipalities with pipelines (and their neighbors) and the rest of the country.
16. See Figure S1 in the Supporting Information.
17. Authors' calculations with data from the Energy Information Administration of the United States (EIA, 2022).
18. SINAIS for its acronym in Spanish.

19. This strategy is akin to the interpretation of turf wars used by Robles et al. (2015) y Jarillo et al. (2016). In those papers, the authors define drug-related turf war exposure when, "the monthly number of firearm-related homicides in a locality surpasses a threshold of two standard deviations above its moving average of the previous four years" (Jarillo et al., (2016), p.138). Although our definition could be considered more static, it is also easier to interpret.
20. Specifically, we aggregate the daily prices of crude oil (WTI spot benchmark) from the EIA.
21. SENER of its acronym in Spanish.
22. See Figures S6 and S7 in the Supporting Information.
23. Formally, our locality-level estimation is:
24. We also tested these models employing different time periods in the data, specifically the initial expansion of the War on Drugs only (2006–2014) and the period after 2000; results remain unchanged. See Table S9 and S10 in the Supporting Information.
25. These results are also consistent with the findings by Alcocer (2022). However, they contradict the findings by Esberg (2022). Moreover, they use different data to measure the presence of criminal organizations. In sum, a deeper understanding of the dynamics between criminal organizations and profits is needed to provide a final conclusion.

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